School of Mathematical Sciences

華東師範大學
EAST CHINA NORMAL UNIVERSITY

Example Workshop in Algebraic Geometry

A handbookformyself demonstration

Friday / 20 November 2026

09:00–15:10 / Shanghai time

East China Normal University

Mathematics Building / Example programme

Illustrative event, speakers and room labels



Photograph: School of Mathematical Sciences, ECNU

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Meeting information

An illustrative programme for a small mathematics workshop

Venue and registration

The programme uses the Mathematics Building at ECNU as its visual setting. Speaker names, times and room labels are examples for adapting the template. Registration opens at 08:40 in the example foyer. **中文示例：**本手册中的讲者、日程及房间名称均为演示内容。

Session format

Each talk has a forty-minute slot, including questions. Morning and afternoon are separate sessions; the programme below gives the order explicitly.

Notes for speakers

Please include the title and speaker name on the first slide. A short board discussion is welcome after the presentation.

Before the session

Check the display connection before your talk. Any change to a title belongs in the final programme, so please send corrections to the organisers before the session begins.

Refreshments

Coffee is served in the example foyer. The midday break gives participants time to eat nearby, and the final discussion brings both sessions together.

Contact and organisers

For this demonstration, use organiser@example.org as the placeholder contact address. Replace it with the organisers' address when preparing a real handbook.

Schedule

Friday / 20 November 2026

Morning

09:00–11:30 / Example lecture room

09:00–09:40 — **Speaker A**, Example Institute — 01 Curves, line bundles and divisors

09:40–10:00 — Coffee and informal discussion

10:00–10:40 — **Speaker B** — 02 Positivity and projective embeddings of X

10:40–10:50 — Interval

10:50–11:30 — **Speaker C**, Example University — 03 Families of projective varieties and the geometric meaning of specialization in moduli problems

11:30–13:00 — Midday break

Afternoon

13:00–15:10 / Example lecture room

13:00–13:40 — **Speaker D**, Example Research Centre — 04 Vector bundles on projective space and a first look at splitting

13:40–14:00 — Interval

14:00–14:40 — **Speaker E**, Example Institute — 05 Pullback, degree growth and elementary examples in algebraic dynamics

14:40–15:10 — Open discussion: questions from the talks and ideas for future collaboration

Abstracts

Morning

01 Curves, line bundles and divisors

Speaker A Example Institute

A divisor on a smooth projective curve determines a line bundle, and sections of that bundle turn geometric questions into linear algebra. We begin with divisors on the projective line and then compare their behaviour on curves of positive genus. If g is the genus of C, the Riemann–Roch formula reads:

$$h^0(C, L) - h^0(C, K_C \otimes L^{-1}) = \deg L + 1 - g.$$

We use the formula to discuss effective divisors and maps to projective space. The examples require only the definitions of a divisor and a line bundle.

02 Positivity and projective embeddings of X

Speaker B

An ample line bundle records how a projective variety can be embedded into projective space after taking a suitable tensor power. We compare ampleness and global generation using line bundles on curves and on projective space. In particular, the bundles $\mathcal{O}_{\mathbb{P}^n}(d)$ give a concrete family in which sections and the associated morphisms can be described explicitly.

The second part follows the same questions through simple constructions: restricting a line bundle to a subvariety, taking tensor products, and pulling back sections along a morphism. These examples separate the geometric role of an embedding from the algebraic description of its coordinate sections.

03 Families of projective varieties and the geometric meaning of specialization in moduli problems

Speaker C Example University

A family organizes several algebraic varieties into a single geometric object. Our running example is the family of plane conics

$$\mathcal{X} = \{x_0 x_1 = t x_2^2\} \subset \mathbb{P}^2 \times \mathbb{A}^1, \qquad \pi : \mathcal{X} \longrightarrow \mathbb{A}^1.$$

For nonzero t the fiber is a smooth conic, while the fiber over zero is the union of two lines. This small example makes specialization visible without requiring a general construction of a moduli space.

We will keep track of two questions throughout the discussion:

1. Which features can be read from a single fiber?
2. Which questions require the total space and the map π?

We finish by comparing different choices of the parameter and explaining why a description of the fibers alone does not describe the whole family.

Afternoon

04 Vector bundles on projective space and a first look at splitting

Speaker D Example Research Centre

On the projective line, every vector bundle is a direct sum of line bundles:

$$E \cong \bigoplus_{i=1}^{r} \mathcal{O}_{\mathbb{P}^1}(a_i).$$

We discuss how the integers a_i describe sections and restrictions, and why the higher-dimensional problem has a different character. The Euler sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$

supplies a familiar bundle with which to test the same questions.

05 Pullback, degree growth and elementary examples in algebraic dynamics

Speaker E Example Institute

Iterating an algebraic map gives a sequence of pullbacks. For an endomorphism $f : \mathbb{P}^k \to \mathbb{P}^k$ of algebraic degree d, the hyperplane class H satisfies $f^*H = dH$ and the iterates have degree d^n. We use this calculation to connect iteration with linear algebra on cohomology.

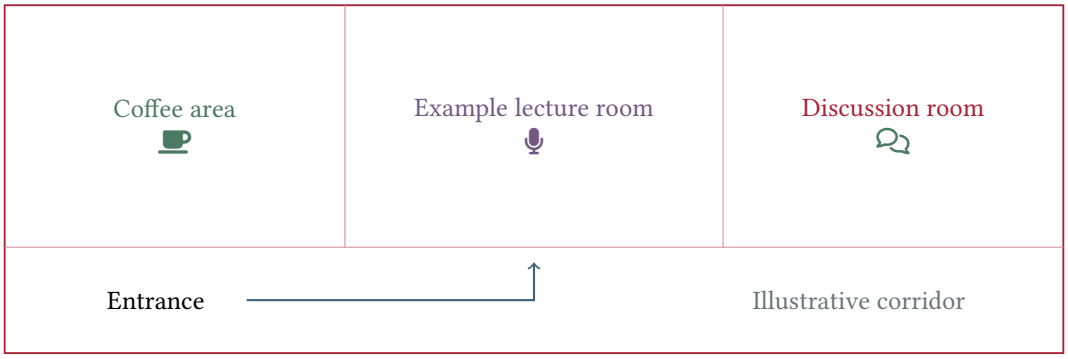
The talk compares these projective-space examples with maps on curves. The aim is to identify which quantities are being measured before moving to more general varieties and rational self-maps. Notation for sheaves, such as $\mathcal{E}$ and $\mathcal{F}$, will be introduced when it is needed.

Logistics

Example arrangements and an illustrative floor plan

Venue plan

The following drawing illustrates how a venue plan can be included directly in the content file. Its room labels and route are demonstration data.



Meals and accommodation

Arrangements to replace

Meals Insert confirmed dining locations and service times.
Accommodation Add the hotel name, address and check-in information.
Transport Add the appropriate station, campus gate and walking route.

Travel checklist

Allow time to reach the meeting room

1. Check the venue address and the campus entrance before departure.
2. Save a local copy of the programme and contact information.
3. Tell the organisers about accessibility or equipment requirements.

Further notes

A free-form section

Discussion questions

Use ordinary paragraphs, lists and equations for material that needs no special box. A new section does not require a new style command or a new data registry.

- Which examples connect two of the talks?
- Which hypotheses are needed for the displayed formulas?
- Which questions should be recorded for a follow-up meeting?

About this example

This booklet demonstrates the public commands of `handbookformyself`. The contents and the schematic plan are original demonstration material. The ECNU logo and building photograph are credited separately in `assets/SOURCES.md`.

> **Adapting the handbook**
>
> Change the cover directly, choose the output mode near the beginning of this file, and replace the report declarations. Then list their IDs in the desired order in the programme and the abstract section.